

BedSAT: Antarctica

Exploring what lies beneath using big data and modelling

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Aims

BedSAT: Antarctica is a modelling approach that combines satellite observations of the ice surface, airborne radar surveys of the bed, and numerical ice flow modelling. It is physically informed by observed bed morphometry statistics which feed into synthetic bedrock generation for ISSM ice-sheet models. With BedSAT we aim to use machine learning to model bed-to-surface transfer from physics-based simulations over these synthetic beds, and to invert the transfer process, so that observed surface expression from real satellite data can be mapped back to bed topography. The goal of this project is to improve the resolution and accuracy of Antarctic bed topography.

1.1 Project Deliverables

- **Paper 1 — ODSA:** Develop a physically informed spectral analysis framework for quantifying Antarctic subglacial landscape characteristics and classifying bed morphology into archetypes for ice sheet modelling;
- **Paper 2 — Bedrock factory:** Design a synthetic bed topography generator for Antarctic landscapes that takes morphometric statistics from ODSA as inputs and produces bed realisations at ice sheet model resolution.
- **Paper 1 — BedSAT:** Develop a modelling approach that combines satellite remote sensing, airborne radar and numerical ice flow modelling to improve the resolution and accuracy of Antarctic bed topography.

1.2 Timeline

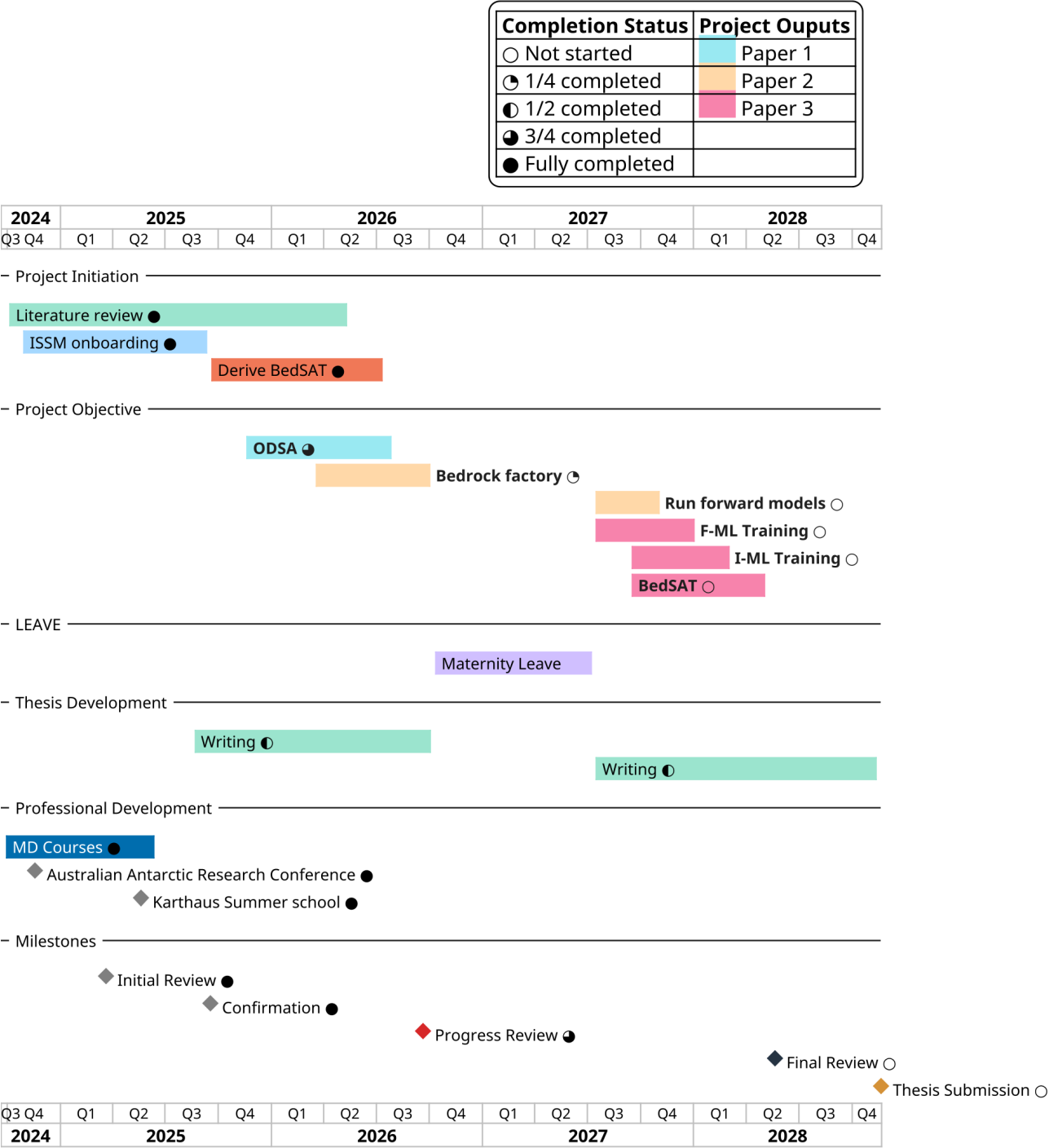


Figure 1.1: Project timeline. In the Project Objective section, the colour of each bar shows which paper the task contributes to. The circle next to each item is an estimate of how complete it is: a quarter-filled circle means that the prototype of the code and workflow exists. A half-filled circle means that the task workflow is in progress. A three-quarter-filled circle means that the results are complete and a paper needs to be written up or the task needs to be officially approved by the panel. A filled circle means that the task is complete.

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Preliminary Literature review

Bed topography is one of the most crucial boundary conditions that influences ice flow and loss from the Antarctic Ice Sheet (AIS) [1]. Bed topography datasets are typically generated from airborne radar echo sounding (RES) surveys, which are sparse and unevenly distributed across the Antarctic continent. Interpolation schemes used to “gap fill” these sparse datasets yield bed topography estimates that have uncertainties of multiple hundreds of metres in elevation [1]. Some examples of these products are the Bedmap ice bed, surface and thickness gridded datasets for Antarctica [2–4], or BedMachine [5,6]. These uncertainties then propagate through simulations of AIS evolution under climate change [7]. Given the logistical challenges of accessing large parts of Antarctica, there is a need for alternative approaches that integrate diverse and possibly more spatially complete data streams, including satellite data.

2.1 Approaches to Bed Topography Reconstruction

To address the gap in understanding subglacial conditions, the research goal of this project is to develop a method that combines forward and inverse modelling, leveraging high-resolution satellite surface data to significantly improve bed topography estimates in regions where direct radar measurements are sparse. The generation of gridded bed topographies in Antarctica is based on two primary modelling approaches: forward and inverse models.

Forward models aim to understand how different bed properties influence ice flow. These often generate interpolated gridded topography using geostatistical techniques like kriging [8] and are valuable for investigating how uncertainties in the bed can impact outcomes like simulated ice mass loss. Forward models can generally be classed as elevation-preserving or texture-preserving (the bed’s geostatistical properties are maintained). Their limitation is the reliance on assumptions about the bed conditions rather than direct observation, so they explore possibilities instead of determining the actual topography.

Inverse models involve methods that assimilate other data sources and work by inferring bed properties, such as topography or basal slipperiness, from surface observations like ice elevation and velocity. Techniques like control method inversion [9] and 4dvar assimilation minimise mismatches between observed and simulated data. On the other hand, mass conservation approaches use physical laws to reconstruct the bed, especially where measurements are sparse [1,5]. Despite their power, these methods face significant challenges. Variational approaches often require regularisation to prevent non-physical results and over-fitting [10], and time-dependent methods like 4dvar are computationally expensive. Probabilistic methods such as Markov Chain Monte Carlo (MCMC) are powerful for quantifying uncertainty but remain too computationally intensive [10] for large-scale models.

Ultimately, the limitations of these methods reveal a core challenge in modern glaciology: the need to develop a new approach that can fully utilise rich satellite surface data to overcome the lack of direct subglacial measurements.

2.2 Quantifying Bed Morphometry

It is still uncommon for methods to systematically quantify the relationship between bed topography and its morphometric landscape statistics. Studies that measure the statistics report them as continuous fields, and terrain types are assigned to specific regions rather than to individual cells, with each parameter described qualitatively. Siegert et al. [11] define three bed roughness groups across named areas of East Antarctica, and Rippin et al. [12] assign three terrain types to the areas of their survey. Numerical thresholds have been published for single parameters, and Jordan et al. [13] divide the Hurst exponent at 0.5 and 0.75 to define low, medium and high roughness categories, but no scheme applies thresholds to a combination of axes per analysed cell. For published classified maps, each cell is often assigned to one class with no margin reported alongside it: Ockenden et al. [14] map the Antarctic bed into five landscape classes on 50 km cells, and their accompanying data release [15] records each assignment without a confidence interval or an alternative class. For classification schemes in general, ambiguity in landscape class is often only acknowledged as a caveat or analysed under a qualitative lens of multiple interpretations but it is not quantified, so a published class cannot be read alongside the alternatives that the measurement might still admit. That absence of margins makes such maps vulnerable to survey artefacts, like local roughness measurement bias due to radar migration status [11, 16, 17] or measurement limitations from the orientation and along-track sampling resolution due to uneven survey coverage [18].

Assigning a single class to a region or cell also averages the terrain within it. Rippin et al. [12] note that their large-scale grouping masks some complexities within the groups, and the selective erosion class of Ockenden et al. [14], which covers 56% of Antarctica, is defined as all topography that is not clearly “low-relief” or “alpine”. Such classes cannot capture the gradual change in roughness reported by Bingham and Siegert [19] or the continuum of landscape types described by Sugden [20].

In chapter 5, we introduce Observational Data Spectral Analysis (ODSA), a physically informed framework for quantifying Antarctic subglacial landscape characteristics and classifying bed morphology into archetypes for ice sheet modelling. The landscape classification provides an empirical foundation for the generation of structurally realistic synthetic bed topographies. ODSA operates on the along-track picks published with Bedmap3 instead of a gridded product, which allows us to measure the survey data directly and avoid the interpolation issues described earlier. The tracks used are visible in Fig. 2.1.

2.3 From Signal Transfer to Tractable Inversion

This project aims to develop a framework to connect observable surface features to the subglacial bed below, ensuring that the governing physics of ice sheet flow are respected. To do this, our work focuses on utilising surface-bed transfer functions.

The core principle behind a transfer function is that the ice sheet acts as a physical filter, modifying the expression of bed features as they propagate to the surface. The work by Budd [21] established that ice preferentially dampens short-wavelength bed undulations while features with a wavelength approximately $3.3\times$ the ice thickness are most clearly expressed at the surface. This process also introduces a phase lag, and can be described mathematically using frequency-dependent transfer functions. More recent studies have built directly on this foundation. For instance, Gudmundsson and Raymond [22] refined the transfer function concept for ice streams, and Ockenden et al. [23] applied it in reverse, using full-Stokes transfer functions to invert high-resolution satellite observations of surface elevation and velocity to infer bed properties. These studies collectively establish that the physical relationship between

the bed and the surface provides a viable pathway for subglacial mapping.

Despite the success of these approaches, their practical application has been limited by simplifying assumptions about ice physics. A primary limitation is the reliance on a linear (Newtonian) ice rheology, where stress is directly proportional to the strain rate (i.e., a “constant viscosity”). This assumption is often utilised to make the inversion mathematically tractable, but it contrasts with the widely accepted non-linear Glen’s Flow Law, where the stress exponent is typically $n \approx 3$ or even $n = 4$. The choice of rheology is a critical control on the bed-to-surface signal transfer; a non-linear rheology ($n = 4$) produces significantly different surface expressions compared to a linear one ($n = 1$), and by largely ignoring non-linear rheology and complex basal sliding conditions, past inversion methods have introduced uncertainties and may not be robustly applicable across all dynamic regimes of the ice sheet.

The opportunity lies in exploiting the vast wealth of underutilised high-resolution satellite surface observations including NASA’s MEaSUREs [24] velocities and REMA elevations [25]. Our approach with BedSAT will harness these data streams by building upon established transfer function theory while addressing the fundamental limitation that has restricted past approaches: the mathematical intractability of non-linear ice physics.

BedSAT will connect surface observations with bed topography using more realistic rheological assumptions ($n = 4$) and complex sliding conditions, rather than the simplified linear physics commonly used by traditional inversions. The key step to making this process tractable is Physics-Informed Machine Learning (Physics-ML), which solves the computational bottleneck that has forced previous methods to rely on unrealistic simplifications. By leveraging NVIDIA PhysicsNeMo, a framework designed to blend governing physics (PDEs) with training data [26], BedSAT can learn the non-linear mapping between surface expressions and bed topography without linearising the physics. This approach would transform what was previously computationally intractable into a fast, accurate inverse solver. Through an iterative process where initially inverted bed topography is integrated into forward models for progressive refinement, BedSAT will deliver physically consistent reconstructions validated against independent datasets. This would represent a major improvement: where previous methods had to choose between physical realism and computational feasibility, Physics-ML can enable both simultaneously.

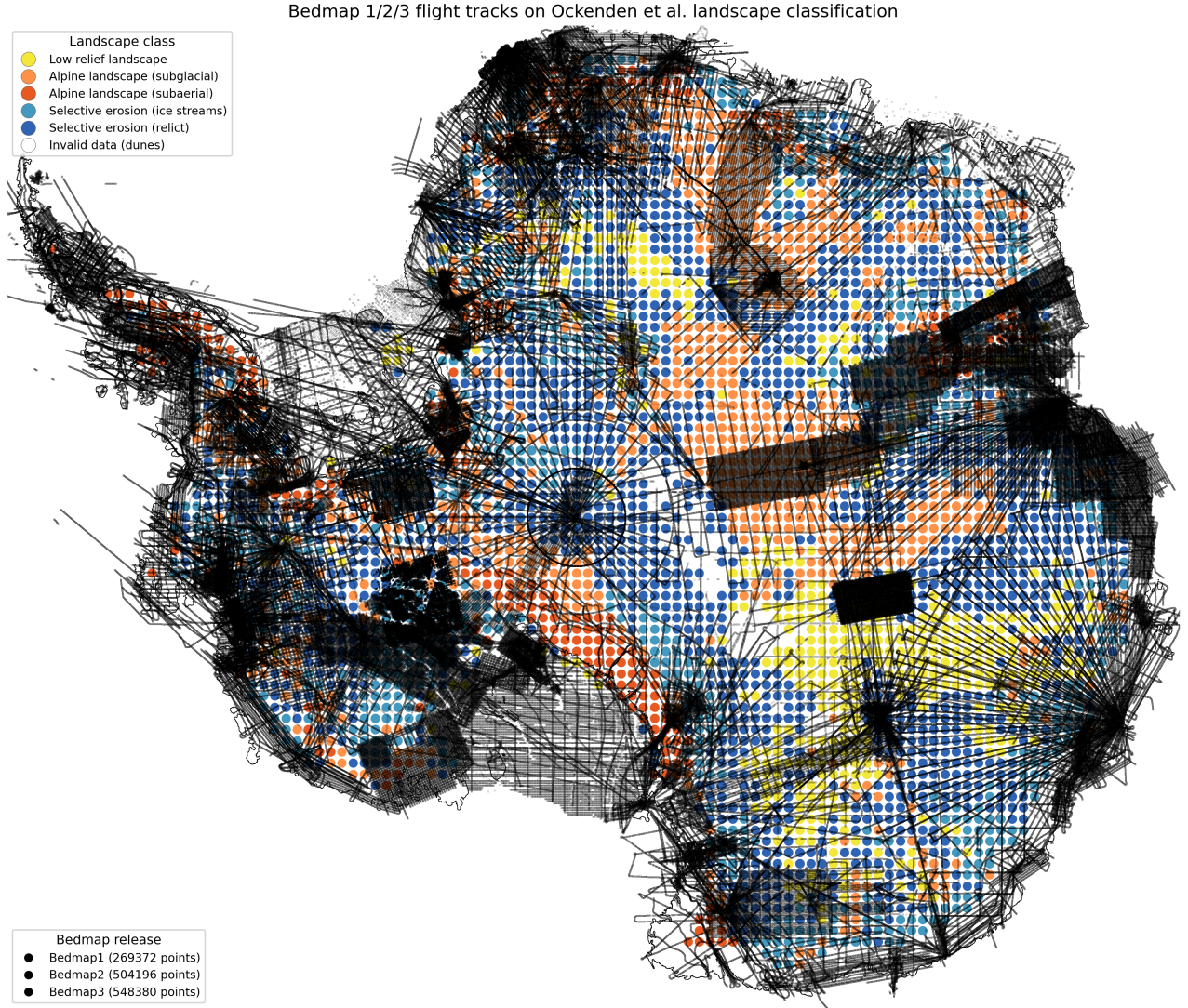


Figure 2.1: Bedmap1, Bedmap2 and Bedmap3 along-track survey points (black) over the continental landscape classification map by Ockenden et al. [14] (coloured cells, cells masked as invalid data (dunes) are drawn white). We reproduce the classification from the published metric grids [15] on 50 km cells. The point counts in the legend show the number of subsampled plotted points rather than published picks. ODSA operates on these along-track data, so the figure shows the sampling available to us across a continental landscape classification.

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Research Significance

The polar regions are losing ice, and their oceans are changing rapidly [27]. The consequences of this extend to the whole planet and it is crucial for us to understand them to be able to evaluate the timing and magnitude of potential mitigation strategies. The accelerating loss of continental ice sheets in both Greenland and Antarctica are major contributors to global sea level rise [27]. Impacts extend beyond direct ice loss: as fresh water from melting ice sheets is added into the ocean, it increases ocean stratification disrupting global thermohaline circulation [28]. In addition, cold freshwater can dissolve larger amounts of CO₂ than regular ocean water creating corrosive conditions for marine life [27]. While there is high confidence in current ice loss and retreat observations in many areas, there is more uncertainty about the mechanisms driving these changes and their future progression [29]. Uncertainty increases in regions with variable bed conditions, where characteristics like bed slipperiness and roughness have been difficult to verify via direct observations. Other problematic areas involve the ice sheet grounding line (GL): The zone that delineates where the ice sheet is grounded on bedrock. The GL retreat rate depends crucially on topographical features like pinning points [29], which lead to increased retardation of upstream flow. Although this mechanism is established, major knowledge gaps persist in mapping bed topography across Antarctic ice sheet margins, with over half of all margin areas having insufficient data within 5km of the grounding zone [30]. Addressing this data gap through systematic mapping enhanced by auxiliary data streams with more complete coverage, would significantly improve both our understanding of current ice dynamics and the accuracy of ice-sheet models projecting future changes.

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Project Methodology

4.1

4.2 Development of the BedSAT Inversion Framework

BedSAT is a model that comprises physically informed transfer function, leveraging an improved understanding of how ice rheology and basal sliding conditions affect the surface expression of subglacial topography. The framework encompasses the following stages.

1. **Training the Machine-Learning model:** The BedSAT framework will be trained and validated using a synthetic bedrock database and a data-rich region, such as the Aurora Subglacial Basin, where extensive ice-penetrating radar data of the bed is available for direct comparison.

Forward Ice Flow Model: The forward model of ice dynamics will be configured for the chosen study site, including key model assumptions, such as the choice of ice flow equations and the parameters governing ice rheology. The robustness of the numerical configuration will be confirmed through grid independence testing to ensure that the results are not an artefact of the model's spatial resolution.

Inversion Pre-processing: The primary observational inputs for the model will be satellite-derived surface velocity, surface elevation [24, 25] and ice thickness [31, 32]. A critical pre-processing step will involve filtering high-frequency noise (e.g., aeolian features) from the surface data to ensure that the glaciological signal from the underlying bed is isolated, this step will require assessment from experts in the field.

2. **Inverse Modelling to Estimate Bed Topography:** The core of the BedSAT framework is the inversion. BedSAT will iteratively adjust an initial estimate of the bed topography, aiming to minimize the mismatch between the forward model's simulated surface velocities and the satellite-observed velocities.
3. **Validation and Sensitivity Analysis:** The framework's performance will be rigorously evaluated. First, the inverted bed topography will be directly validated against the "ground-truth" radar-derived bed maps for the study region. Second, a series of sensitivity analyses will be conducted to quantify the model's robustness. These tests will systematically alter key assumptions, such as the rheological parameters for different regions, to determine their impact on the final inverted topography and to establish the uncertainty bounds of the framework's results.

4.3 BedSAT: Antarctica - Derive a new bed topography

I will apply the validated BedSAT methodology from Objective 1 to the entire Antarctic continent, deriving a new continent-wide bed topography dataset. Using covariance properties from

existing radar surveys, I will generate multiple realisations of the bed, each with unique and statistically-consistent topographic roughness

4.4 Evaluate the impact of new bed topography

The new bed topography datasets will be used to conduct a sensitivity analysis of ice sheet model projections to 2300CE. This will investigate the impact of the improved topography and different roughness realisations on ice dynamics, subglacial hydrology, and overall ice mass loss from Antarctica, directly addressing the project's main research questions.

411 words in this section. PM should be (200-500 words)

Progress

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Bibliography

- [1] M. Morlighem, E. Rignot, T. Binder, D. Blankenship, R. Drews, G. Eagles, O. Eisen, F. Ferraccioli, R. Forsberg, P. Fretwell, V. Goel, J. S. Greenbaum, H. Gudmundsson, J. Guo, V. Helm, C. Hofstede, I. Howat, A. Humbert, W. Jokat, N. B. Karlsson, W. L. Lee, K. Matsuoka, R. Millan, J. Mouginot, J. Paden, F. Pattyn, J. Roberts, S. Rosier, A. Ruppel, H. Seroussi, E. C. Smith, D. Steinhage, B. Sun, M. R. van den Broeke, T. D. van Ommen, M. van Wessem, and D. A. Young. *Deep glacial troughs and stabilizing ridges unveiled beneath the margins of the Antarctic ice sheet*. Nature Geoscience **13**, 132 (2020). DOI: [10.1038/s41561-019-0510-8](https://doi.org/10.1038/s41561-019-0510-8). 6
- [2] M. B. Lythe and D. G. Vaughan. *BEDMAP: A new ice thickness and subglacial topographic model of Antarctica*. Journal of Geophysical Research: Solid Earth **106**, 11335 (2001). DOI: [10.1029/2000JB900449](https://doi.org/10.1029/2000JB900449). 6
- [3] P. Fretwell, H. D. Pritchard, D. G. Vaughan, J. L. Bamber, N. E. Barrand, R. Bell, C. Bianchi, R. G. Bingham, D. D. Blankenship, G. Casassa, G. Catania, D. Callens, H. Conway, A. J. Cook, H. F. J. Corr, D. Damaske, V. Damm, F. Ferraccioli, R. Forsberg, S. Fujita, Y. Gim, P. Gogineni, J. A. Griggs, R. C. A. Hindmarsh, P. Holmlund, J. W. Holt, R. W. Jacobel, A. Jenkins, W. Jokat, T. Jordan, E. C. King, J. Kohler, W. Krabill, M. Riger-Kusk, K. A. Langley, G. Leitchenkov, C. Leuschen, B. P. Luyendyk, K. Matsuoka, J. Mouginot, F. O. Nitsche, Y. Nogi, O. A. Nost, S. V. Popov, E. Rignot, D. M. Rippin, A. Rivera, J. Roberts, N. Ross, M. J. Siegert, A. M. Smith, D. Steinhage, M. Studinger, B. Sun, B. K. Tinto, B. C. Welch, D. Wilson, D. A. Young, C. Xiangbin, and A. Zirizzotti. *Bedmap2: improved ice bed, surface and thickness datasets for Antarctica*. The Cryosphere **7**, 375 (2013). DOI: [10.5194/tc-7-375-2013](https://doi.org/10.5194/tc-7-375-2013). 6
- [4] H. D. Pritchard, P. T. Fretwell, A. C. Fremand, J. A. Bodart, J. D. Kirkham, A. Aitken, J. Bamber, R. Bell, C. Bianchi, R. G. Bingham, D. D. Blankenship, G. Casassa, K. Christianson, H. Conway, H. F. J. Corr, X. Cui, D. Damaske, V. Damm, B. Dorschel, R. Drews, G. Eagles, O. Eisen, H. Eisermann, F. Ferraccioli, E. Field, R. Forsberg, S. Franke, V. Goel, S. P. Gogineni, J. Greenbaum, B. Hills, R. C. A. Hindmarsh, A. O. Hoffman, N. Holschuh, J. W. Holt, A. Humbert, R. W. Jacobel, D. Jansen, A. Jenkins, W. Jokat, L. Jong, T. A. Jordan, E. C. King, J. Kohler, W. Krabill, J. Maton, M. K. Gillespie, K. Langley, J. Lee, G. Leitchenkov, C. Leuschen, B. Luyendyk, J. A. MacGregor, E. MacKie, G. Moholdt, K. Matsuoka, M. Morlighem, J. Mouginot, F. O. Nitsche, O. A. Nost, J. Paden, F. Pattyn, S. Popov, E. Rignot, D. M. Rippin, A. Rivera, J. L. Roberts, N. Ross, A. Ruppel, D. M. Schroeder, M. J. Siegert, A. M. Smith, D. Steinhage, M. Studinger, B. Sun, I. Tabacco, K. J. Tinto, S. Urbini, D. G. Vaughan, D. S. Wilson, D. A. Young, and A. Zirizzotti. *Bedmap3 updated ice bed, surface and thickness gridded datasets for Antarctica*. Scientific Data **12**, 414 (2025). DOI: [10.1038/s41597-025-04672-y](https://doi.org/10.1038/s41597-025-04672-y). 6
- [5] M. Morlighem, C. N. Williams, E. Rignot, L. An, J. E. Arndt, J. L. Bamber, G. Catania, N. Chauché, J. A. Dowdeswell, B. Dorschel, I. Fenty, K. Hogan, I. Howat, A. Hubbard,

- M. Jakobsson, T. M. Jordan, K. K. Kjeldsen, R. Millan, L. Mayer, J. Mouginot, B. P. Y. Noël, C. O’Cofaigh, S. Palmer, S. Rysgaard, H. Seroussi, M. J. Siegert, P. Slabon, F. Straneo, M. R. van den Broeke, W. Weinrebe, M. Wood, and K. B. Zinglarsen. *Bedmachine v3: Complete bed topography and ocean bathymetry mapping of greenland from multibeam echo sounding combined with mass conservation*. *Geophysical Research Letters* **44**, 11,051 (2017). ARXIV: [10.1002/2017GL074954](https://arxiv.org/abs/10.1002/2017GL074954), DOI: [10.1002/2017GL074954](https://doi.org/10.1002/2017GL074954). 6
- [6] M. Morlighem, R. Charrassin, N. Maffezzoli, R. Millan, H. Ockenden, N.-J. Schlegel, H. Seroussi, and M. Wood. *New high-resolution subglacial bed topography and coastal bathymetry of the Antarctic and Greenland Ice Sheets*. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* **384**, 20240538 (2026). DOI: [10.1098/rsta.2024.0538](https://doi.org/10.1098/rsta.2024.0538). 6
- [7] B. A. Castleman, N. J. Schlegel, L. Caron, E. Larour, and A. Khazendar. *Derivation of bedrock topography measurement requirements for the reduction of uncertainty in ice-sheet model projections of Thwaites Glacier*. *The Cryosphere* **16**, 761 (2022). DOI: [10.5194/tc-16-761-2022](https://doi.org/10.5194/tc-16-761-2022). 6
- [8] E. J. MacKie, D. M. S. J. Caers, M. R. Siegfried, and C. Scheidt. *Antarctic Topographic Realizations and Geostatistical Modeling Used to Map Subglacial Lakes*. *Journal of Geophysical Research: Earth Surface* **125** (2020). DOI: [10.1029/2019JF005420](https://doi.org/10.1029/2019JF005420). 6
- [9] J. De Rydt, G. H. Gudmundsson, H. F. J. Corr, and P. Christoffersen. *Surface undulations of Antarctic ice streams tightly controlled by bedrock topography*. *The Cryosphere* **7**, 407 (2013). DOI: [10.5194/tc-7-407-2013](https://doi.org/10.5194/tc-7-407-2013). 6
- [10] M. Morlighem and D. Goldberg. *Data Assimilation in Glaciology*, pages 93–111. Cambridge University Press (2024). DOI: [10.1017/9781009180412.007](https://doi.org/10.1017/9781009180412.007). 6
- [11] M. J. Siegert, J. Taylor, and A. J. Payne. *Spectral roughness of subglacial topography and implications for former ice-sheet dynamics in East Antarctica*. *Global and Planetary Change* **45**, 249 (2004). DOI: [10.1016/j.gloplacha.2004.09.008](https://doi.org/10.1016/j.gloplacha.2004.09.008). 7
- [12] D. M. Rippin, R. G. Bingham, T. A. Jordan, A. P. Wright, N. Ross, H. F. J. Corr, F. Ferraccioli, A. M. L. Brocq, K. C. Rose, and M. J. Siegert. *Basal roughness of the Institute and Möller Ice Streams, West Antarctica: process determination and landscape interpretation*. *Geomorphology* **214**, 139 (2014). DOI: [10.1016/j.geomorph.2014.01.021](https://doi.org/10.1016/j.geomorph.2014.01.021). 7
- [13] T. M. Jordan, M. A. Cooper, D. M. Schroeder, C. N. Williams, J. D. Paden, M. J. Siegert, and J. L. Bamber. *Self-affine subglacial roughness: consequences for radar scattering and basal water discrimination in northern Greenland*. *The Cryosphere* **11**, 1247 (2017). DOI: [10.5194/tc-11-1247-2017](https://doi.org/10.5194/tc-11-1247-2017). 7
- [14] H. Ockenden, R. G. Bingham, D. Goldberg, A. Curtis, and M. Morlighem. *Complex mesoscale landscapes beneath Antarctica mapped from space*. *Science* **391**, 314 (2026). DOI: [10.1126/science.ady2532](https://doi.org/10.1126/science.ady2532). 7, 9
- [15] H. Ockenden. *Data accompanying manuscript: Complex mesoscale landscapes beneath Antarctica mapped from space*, (2025). DOI: [10.5281/zenodo.17226363](https://doi.org/10.5281/zenodo.17226363). 7, 9
- [16] J. Taylor, M. J. Siegert, A. J. Payne, and B. Hubbard. *Regional-scale bed roughness beneath ice masses: measurement and analysis*. *Computers & Geosciences* **30**, 899 (2004). DOI: [10.1016/j.cageo.2004.06.007](https://doi.org/10.1016/j.cageo.2004.06.007). 7

- [17] D. M. Rippin. *The basal roughness of Pine Island Glacier, West Antarctica*. Journal of Glaciology **57**, 67 (2011). DOI: [10.3189/002214311795306574](https://doi.org/10.3189/002214311795306574). 7
- [18] R. G. Bingham and M. J. Siegert. *Quantifying subglacial bed roughness in Antarctica: implications for ice-sheet dynamics and history*. Quaternary Science Reviews **28**, 223 (2009). DOI: [10.1016/j.quascirev.2008.10.014](https://doi.org/10.1016/j.quascirev.2008.10.014). 7
- [19] R. G. Bingham and M. J. Siegert. *Radar-derived bed roughness characterization of Institute and Möller ice streams, West Antarctica, and comparison with Siple Coast ice streams*. Geophysical Research Letters **34**, L21504 (2007). DOI: [10.1029/2007GL031483](https://doi.org/10.1029/2007GL031483). 7
- [20] D. E. Sugden. *Glacial erosion by the Laurentide ice sheet*. Journal of Glaciology **20**, 367 (1978). DOI: [10.3189/S0022143000013915](https://doi.org/10.3189/S0022143000013915). 7
- [21] W. F. Budd. *Ice Flow Over Bedrock Perturbations*. Journal of Glaciology **9**, 29 (1970). DOI: [10.3189/S0022143000026770](https://doi.org/10.3189/S0022143000026770). 7
- [22] G. H. Gudmundsson and M. Raymond. *On the limit to resolution and information on basal properties obtainable from surface data on ice streams*. The Cryosphere **2**, 167 (2008). DOI: [10.5194/tc-2-167-2008](https://doi.org/10.5194/tc-2-167-2008). 7
- [23] H. Ockenden, R. G. Bingham, A. Curtis, and D. Goldberg. *Ice-flow perturbation analysis: a method to estimate ice-sheet bed topography and conditions from surface datasets*. Journal of Glaciology **69**, 1677 (2023). DOI: [10.1017/jog.2023.50](https://doi.org/10.1017/jog.2023.50). 7
- [24] A. S. Gardner, M. A. Fahnestock, and T. A. Scambos. *ITS_LIVE Regional Glacier and Ice Sheet Surface Velocities: Version 1*, (2019). Data archived at National Snow and Ice Data Center. DOI: [10.5067/6II6VW8LLWJ7](https://doi.org/10.5067/6II6VW8LLWJ7). 8, 13
- [25] I. M. Howat, C. Porter, B. E. Smith, M.-J. Noh, and P. Morin. *The Reference Elevation Model of Antarctica*, (2019). The Cryosphere. DOI: [10.5194/tc-13-665-2019](https://doi.org/10.5194/tc-13-665-2019). 8, 13
- [26] NVIDIA. *NVIDIA PhysicsNeMo*. <https://developer.nvidia.com/physicsnemo>, (2025). Accessed: 2025-09-21. 8
- [27] M. Meredith, M. Sommerkorn, S. Cassotta, C. Derksen, A. Ekaykin, A. Hollowed, G. Kofinas, A. Mackintosh, J. Melbourne-Thomas, M. M. C. Muelbert, G. Ottersen, H. Pritchard, and E. A. G. Schuur. *The Ocean and Cryosphere in a changing climate*. In H.-O. Pörtner, D. C. Roberts, V. Masson-Delmotte, P. Zhai, M. Tignor, E. Poloczanska, K. Mintenbeck, A. Alegría, M. Nicolai, A. Okem, and J. Petzold (editors), *IPCC Special Report on the Ocean and Cryosphere in a Changing Climate*, pages 203—320. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA (2019). DOI: [10.1017/9781009157964.005](https://doi.org/10.1017/9781009157964.005). 11
- [28] S. S. Jacobs. *Bottom water production and its links with the thermohaline circulation*. Antarctic Science **16**, 427–437 (2004). DOI: [10.1017/S095410200400224X](https://doi.org/10.1017/S095410200400224X). 11
- [29] B. Fox-Kemper, H. T. Hewitt, C. Xiao, G. Aðalgeirsdóttir, S. S. Drijfhout, T. L. Edwards, N. R. Golledge, M. Hemer, R. E. Kopp, G. Krinner, A. Mix, D. Notz, S. Nowicki, I. S. Nurhati, L. Ruiz, J.-B. Sallée, A. B. A. Slangen, and Y. Yu. *Ocean, Cryosphere and Sea Level Change*. In V. Masson-Delmotte, P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J. B. R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (editors), *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the*

- Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, chapter 9, pages 1211–1362. Cambridge University Press, Cambridge, UK and New York, NY, USA (2021). DOI: [10.1017/9781009157896.011](https://doi.org/10.1017/9781009157896.011). 11
- [30] R. A. Group. *RINGS: Collaborative international effort to map all Antarctic ice-sheet margins*. Scientific Committee on Antarctic Research (2022). DOI: [10.5281/zenodo.6638327](https://doi.org/10.5281/zenodo.6638327). 11
- [31] D. A. Young, A. P. Wright, J. L. Roberts, R. C. Warner, N. W. Young, J. S. Greenbaum, D. M. Schroeder, J. W. Holt, D. E. Sugden, D. D. Blankenship, and T. D. van Ommen. *A dynamic early East Antarctic Ice Sheet suggested by ice-covered fjord landscapes*. *Nature* **474**, 72 (2011). DOI: [10.1038/nature10114](https://doi.org/10.1038/nature10114). 13
- [32] Australian Antarctic Division. Australian Antarctic Program: Project 4077 - Sea Ice Thickness Measurements [online. (2013), cited 2025-02-12]. 13